NDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत

Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

Mid Sem-II | **Subject: Digital Electronics (ECL-202)** | **Roll no** 12412003

Branch: CSE/IT | **M.M.-15 marks** | **Time- 60 mint.**

Note: All questions are compulsory.

Q-1	A combinational circuit is defined by the following two Boolean functions: $F1 = x'y'z' + xz$ and $F2 = \sum(0,2,4,7)$ Design the combinational circuit that implements the functions with a decoder and external gates.	(3 marks)
Q-2	Design a security access control system using a 4×1 multiplexer where only the binary input combinations 011, 101 and 110 should grant access. Show the circuit and find the Boolean expression using multiplexer.	(3 marks)
Q-3	Design a characteristics and excitation tables for a flip-flop with A and B as inputs with given characteristics equation. $Q_{(n+1)} = A'Q' + B'Q$	(3 marks)
Q-4	Design a 3- bit synchronous down counter using T flip flops.	(3 marks)
Q-5	Write a short note on optical disk memory and static RAM.	(3 marks)

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Mid Sem-II
Branch: CSE/IT/DSA M.-15 marks
Semester II___2024-28 Batch

Subject: COMMUNICATION SKILLS (HUL 204)
Time- 60 min.
Roll No. ......12412003..........

Attempt All. (3 X 5 = 15)

1. Write a note on different types of reading. With the help of suitable examples, contextualize them according to the purpose of reading.

2. What do you understand by the Thesis statement and the topic sentence? With the help of suitable examples, explain the different types of Thesis statements. Also, comment on the significance of the topic sentence and the thesis statement in writing.

3. Write a paragraph in the Narrative style of writing. Then, rewrite the same paragraph in the descriptive style of writing.

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2nd Semester Mid Sem-II | **Subject: Web Designing (UCS 003)** | **Roll no:** 12412003

Branch: CSE/IT | **M.M.-15 marks** | **Time- 60 minutes**

Note: All questions are compulsory.

Q-1	a. Imagine you're explaining JavaScript to a friend who has only used HTML and CSS. Describe what JavaScript adds to a web page and why it's important. (1 mark) b. Explain the difference between the loose equality operator (==) and the strict equality operator (===) in JavaScript. Include a code example that highlights their differing behaviors. (1 mark) c. Write a snippet of JavaScript that outputs the text "Welcome to JS!" into the browser console. (1 mark)
Q-2	a. What is the difference between declaring a function using the function keyword and assigning it to a variable as an anonymous function? Provide code examples for both. (2 marks) b. JavaScript uses something called "scope." What happens if you declare a variable inside a function and try to use it outside the function? (1 mark) c. Why do we use comments in our code? Provide both a single-line and a multi-line comment example in JavaScript. (1 mark) d. Describe how indexOf() and lastIndexOf() string methods work. Write a short code example that finds the first and the last occurrence of the substring "JS" in the string "Learn JS with JS exercises". (1 mark)
Q-3	a. Explain the two categories of argument passing in JavaScript functions. Which data types fall into each category? (1 marks) b. Write a function to generate random integer between 15 & 30 (inclusive) and logs it to console. (1 mark)
Q-4	Assume the following HTML exists on the page: <button id="add-item">Add Item</button> <ul id="todo-list"> <form id="signup"> <input type="text" id="email" name="email" placeholder="you@example.com"> <button type="submit">Submit</button> </form> a. Demonstrate three different ways to select the <ul id="todo-list"> element in JavaScript (e.g. by ID, by tag name, by selector). Show a code snippet for each. (2 marks) b. Write a script that listens for clicks on the "Add Item" button and appends a new with the text "New Task" to the end of the to-do list each time it's clicked. (2 marks) c. Validate the email on form submission; if it lacks "@", stop submission and alert "Please enter a valid email." (1 mark)

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Mid Sem-II	**Subject: OOPS (UCS004)**	**Roll no ..12412023**
Branch: CSE/IT (2nd SEM)	**M.M.-15 marks**	**Time- 60 mint.**

Note: All questions are compulsory.

Please read the question carefully before writing the answer.

Sr. No	Questions	Marks
Q-1.	IIIT Sonepat needs a library management system that reads and writes book details (title, author, ISBN) from/to a file, allows users to search for a book by ISBN, and handles exceptions for cases like: • *File not found* • *ISBN not found in the list* • *Invalid input format* As a software developer, use OOPs concepts (classes, objects), file streams for reading data, templates to store book records, and exception handling to manage errors gracefully.	4
Q-2:	A company has different types of employees – Full-time, Part-time, and Contract-based. Each type has a different salary calculation method: • *Full-time employees get a fixed salary.* • *Part-time employees are paid based on hours worked.* • *Contract-based employees get a fixed amount plus bonuses.* As a System Design Engineer, you have to design a C++ program using inheritance and polymorphism to manage employee payroll efficiently, allowing the company to calculate salaries for any employee type without modifying existing code when new employee types are added.	4
Q-3:	You are hired as a software developer by a newly build hospital. Hospital needs to develop a software system that must have following features, *1). Registers patients and doctors,* *2). Schedules appointments,* *3). Handles different billing policies for each department,* *4). Keeps medical history confidential,* *5). Automatically calculates total bill based on services.* Apply C++ OOPs concept to develop a software that meets the hospital requirement.	7

Roll No: | 1 | 2 | 4 | 1 | 2 | 0 | 0 | 3 |

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B.Tech II Sem II Mid Semester Examination, 23th April, 2025

Mid Sem-II	Subject: Engineering Mathematics-II	Course Code: MAL-201
Branch: IT	Maximum Marks. -15 marks	Time- 60 min.

Note: All questions are compulsory.

Q-1	Find the Complete solution of ODE $(D^2+1)y = x^2 \sin 2x$.	(3marks)
Q-2	Solve the following linear differential equations for $x(t)$ & $y(t)$ $(3D^2+3D)x = 4t-3$, $(D-1)x - D^2y = t^2$ and Also check How many correct number of constant $x(t)$ & $y(t)$ are contain.	(3marks)
Q-3	Reduce the following Partial Differential Equation into a Standard Form-III And Solve it completely $z(p^2-q^2) = x-y$, where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.	(3marks)
Q-4	Applying the Charpit's Method, Solve the following PDE $2zx - px^2 - 2qxy + pq = 0$.	(3marks)
Q-5	Find the Complementary function and Particular integral of the given PDE $(D^2 - DD' + D' - 1)z = \cos(x+2y) + e^y$.	(3marks)

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Mid Sem-II **Subject Name: Data Structure** **Subject code: UCS0002** **Roll no** 12412003
Branch: IT **2nd Semester** **M.M.-15 marks** **Time- 60 min.**

Note: (1) Attempt all the questions.
(2) All questions carry equal marks (5*3=15)

Q-1	Design a queue using two stacks. Ensure that enqueue and dequeue operations behave exactly like a queue. Analyze which implementation (push-heavy or pop-heavy) offers better average performance.	**(5 marks)**
Q-2	i) Given two singly linked lists that may merge at some point to form a Y-shaped structure, find the node at which they merge. Use optimized methods with constant space and linear time. ii) Given a singly linked list L, design an efficient algorithm to detect whether a cycle exists in the list. If a cycle is detected, modify the list to remove the cycle without deleting or losing any node. Clearly explain your approach, the algorithm used for detection and removal, and analyze its time and space complexity.	**(5 marks)**
Q-3	What is an AVL tree, and how does it offer advantages over a Binary Search Tree (BST)? Design Binary search tree by using following sequence: Pre-order traversal: 1 2 4 5 3 6 Post-order traversal: 4 5 2 6 3 1	**(5 marks)**